TestFlow Revenue Architecture

Subscription Plans & Revenue Models

Date 2026-05-30
Author Parakh Sharma (solo founder)
Status Draft for review
Related `PRICING.md`, `SALES_TARGETS.md`, financial-plan `.tex`

1. Why this document

`PRICING.md` commits to one sound idea: a **flat per-workspace annual fee**, no per-seat metering. Keep that as the *spine*. But a single flat fee leaves money — and customers — on the table at both ends of the market: the seasonal two-substation contractor who will never sign an annual contract, and the utility that wants a framework license for 40 sites.

This document layers **seven revenue models** onto that spine so every buyer has a way in and every customer has a way to spend more over time. It is grounded in what the database already supports: `subscriptions.plan_id` (free-text SKU), `subscriptions.seat_count`, `companies.features` (JSONB feature flags read by `has_feature()` / `useFeature()`), and the Razorpay webhook (`upsert_subscription`).

> **Design principle — Land, Expand, Monetize the edges.** The annual subscription is the *land*. Metered add-ons and seat-free capacity tiers are the *expand*. Consumption packs, outcome pricing, white-label, and utility framework deals *monetize the edges* of the market the flat fee cannot reach.

2. The revenue stack at a glance

#	Revenue model	What the customer buys	Recurring?
1	Core subscription ladder	Annual workspace access & tier features	Yes (annual)
2	Hybrid metered add-ons	Usage beyond fair-use (AI, storage, comms)	Yes (variable)
3	Consumption / project packs	Prepaid "project credits", no contract	No (repeat)
4	Outcome / value pricing	Per signed handover / commissioned bay	Yes (variable)
5	Channel & white-label	OEM/EPC resells to subcontractors	Yes (wholesale)
6	Utility framework license	Multi-site, multi-year, on-prem option	Yes (multi-yr)
7	Services & data products	Onboarding, migration, benchmarks, training	Mixed

A customer typically starts in one model and accretes others. A mid-size contractor lands on `team_annual` (1), buys AI-report overages (2), takes a white-label add-on (7), and three years later signs a utility framework (6) when they win a PowerGrid package.

3. Model 1 — Core subscription ladder

The spine. Five rungs instead of the current four, adding a downmarket `starter_annual` so small contractors are not forced to choose between "₹ 2.4L" and "nothing".

Tier	plan_id	Annual (INR)	Active projects	Headline gating
Free trial	`trial`	₹ 0 / 14 days	1	All templates, mobile, no AI
Starter	`starter_annual`	₹ 60,000	3	Mobile + reports; no AI, no SSO
Team	`team_annual`	₹ 2,40,000	unlimited	+ AI reports, audit-log UI, SLA 1 d
Business	`business_annual`	₹ 6,00,000	unlimited	+ custom templates, SSO, 4 h SLA
Enterprise	`enterprise_custom`	₹ 12,00,000+	unlimited	+ CSM, on-prem, custom RBAC

Unchanged rules from `PRICING.md`: priced per workspace, not per seat; annual upfront is default; multi-tenancy, mobile, soft-delete and export are *never* gated. `starter_annual` deliberately excludes AI and SSO so it cannot cannibalise Team.

3.1 Billing cadence as a price lever

Cadence is sold, not given. Each step toward upfront cash earns the customer a discount and earns you working capital.

Cadence	Adjustment	Notes
Annual upfront	list price	Default. Best margin + float.
Quarterly	**+5%**	Paid concession for cash-flow-shy buyers.
Monthly bridge	+0%, 90 days max	Trial→annual ramp only, never permanent.
2-year prepaid	**−20%**	Locks logo + reduces churn risk.
3-year prepaid	**−30%**	Floor; never discount deeper without a concession.

4. Model 2 — Hybrid metered add-ons

Flat platform fee *plus* a meter on the few resources whose cost scales with heavy use. This is the single biggest change to the revenue shape: it turns power users into more revenue without raising the headline price that wins deals.

Metered SKU	plan_id suffix	Unit price	Fair-use included
AI handover reports	ai_report	₹ 500 / report	50/mo Team, unltd Business+
Document retention	storage_gb	₹ 100 / GB-yr	25 GB Team, 100 GB Business
SMS/WhatsApp alerts	comms_pack	₹ 1 / message	email always free
Extra workspace	extra_ws	₹ 90,000 / yr	1 per contract

Mechanics. Meters are counted against the `rate_limits` table that already backs `rate_limit_check`. AI reports are a natural meter — each generation is already concurrency-locked and rate-limited (10/hr). At month close, sum overages above fair-use and raise a Razorpay invoice line. Crucially, **fair-use is generous** so the meter is invisible to 90% of customers and only bites genuine power users — the ones who can afford it.

Guardrail. Never meter a *measurement*. Metering test records or equipment instances makes engineers under-report to dodge the meter — it corrupts the data you exist to protect. Meter only outputs (reports, storage, notifications) and capacity (workspaces), never field input.

5. Model 3 — Consumption / project packs

For the seasonal contractor who refuses an annual commitment: sell prepaid **project credits**. One credit unlocks one project workspace (full Team features) for the life of that commissioning job. Credits never expire.

Pack	plan_id	Credits	Price (INR)	Per project
Single	pack_1	1	₹ 35,000	₹ 35,000
Crew	pack_5	5	₹ 1,50,000	₹ 30,000
Fleet	pack_20	20	₹ 5,00,000	₹ 25,000

Why it works. It converts the "I'll wait until next month to start" objection into a purchase today, and it is a natural *on-ramp to annual*: a contractor who burns 8+ credits a year is shown the math — "you have spent ₹ 2.0L on packs; `team_annual` is ₹ 2.4L for unlimited." The pack price per project is deliberately *higher* than the annual-implied rate so the upgrade always looks rational.

Schema fit. Credits live as a balance in `companies.features` (e.g. `project_credits: 5`); generating a project decrements it. No new table required for v1.

6. Model 4 — Outcome / value pricing (selective)

Tie a slice of price to the value event itself: a **signed, approved handover**. Offered only where a buyer resists fixed fees and wants "pay-for-results". Two shapes:

- **Per commissioned bay** — ₹ 1,500 per equipment instance that reaches `APPROVED`. Predictable for the buyer, scales with their real workload. Cap it at the Team annual so it can never exceed the flat alternative — the cap is the upsell.
- **Per handover report delivered** — ₹ 2,000 per `CLOSED` project's final report. Aligns spend with the deliverable the client actually pays *them* for.

Use sparingly. Outcome pricing maximises trust on the first deal but makes revenue lumpy and harder to forecast; migrate these accounts to flat annual once they trust the product (typically renewal 1).

7. Model 5 — Channel & white-label (B2B2B)

Let EPC majors and switchgear OEMs resell TestFlow to *their* subcontractors under their own brand. You sell wholesale; they own the relationship.

Lever	Terms	Rationale
Wholesale rate	list × **0.6**	Partner keeps the margin + does the selling.
White-label add-on	₹ 1,00,000 / yr	Custom domain, logo, colours (`white_label`).
Minimum commit	10 workspaces / yr	Filters out tyre-kickers; guarantees volume.
Co-marketing	case study + logo rights	Lowers your CAC on the next direct deal.

Schema fit. Each reseller is a parent `company` record; sub-tenants carry `partner_wholesale` as `plan_id` and inherit the `white_label` feature flag. Razorpay bills the *partner* a single consolidated invoice, not each sub-tenant. This is the highest-leverage channel because one signed OEM can deliver 20–50 workspaces with near-zero direct CAC.

8. Model 6 — Utility framework licensing

The top of the market: PowerGrid, NTPC, state transcos. They do not buy "a workspace" — they buy a **framework agreement** covering many sites for several years, often with on-prem or private-cloud requirements.

- **Site-block license** — published Enterprise rate × 0.7 for 20+ sites bought together (matches `SALES_TARGETS` volume play).
- **Multi-year** — 3–5 year terms, annual escalation clause (CPI or flat 5%/yr) written in.
- **On-prem / private cloud** — premium of ₹ 5,00,000+/yr for a dedicated Supabase project or self-hosted deployment; gated by the `on_prem` flag and a separate deployment SKU.
- **Contractual SLA & security review** — priced into the deal, not given away (each costs real founder/engineer time).

These are slow (6–12 month sales cycles) but anchor ARR and create reference logos that close everything below them.

9. Model 7 — Services & data products

Non-subscription revenue that improves margin and stickiness.

Item	Price (INR)	Type
Onboarding / setup fee	₹ 25,000 one-time	Team & Business; waivable as "discount"
Data migration / import	₹ 25,000 one-time	Legacy Excel → TestFlow
Custom report template	₹ 50,000 each	Per template (`custom_templates`)
On-site training (8 h)	₹ 50,000 + travel	Per session
Premium support (4 h resp.)	₹ 2,00,000 / yr	Upsell below Business SLA
Benchmark / analytics pack	*future*	Anonymised, opt-in failure/throughput data

The *data product* is the long-game option: with enough tenants, anonymised, opt-in commissioning benchmarks (mean time-to-handover, failure rates by equipment type) become a sellable industry

report — a revenue stream with *zero* marginal delivery cost. Strictly opt-in; multi-tenancy isolation (RLS) is never compromised for it.

10. Feature-gating matrix (maps to `companies.features`)

Every gate below is a boolean key in the `companies.features` JSONB, enforced by `has_feature()` server-side and `useFeature()` in the UI. Flags default `TRUE` unless a downgrade sets them false — so a missing flag never accidentally locks a paying customer out.

Feature flag	Starter	Team	Business	Enterprise
`ai_reports`	–	✓	✓	✓
`audit_log_ui`	–	✓	✓	✓
`custom_templates`	–	–	✓	✓
`sso`	–	–	✓	✓
`white_label`	–	add-on	add-on	✓
`priority_support`	–	–	✓	✓
`advanced_analytics`	–	–	–	✓
`on_prem`	–	–	–	✓

Never gated (in every tier): all test templates, mobile app, multi-tenant RLS isolation, real-time / 30 s polling sync, soft-delete & restore, PDF + Excel export. Audit-log *capture* is always on (triggers); only the *viewer UI* is gated.

11. Expansion levers & the metric that matters

A flat annual fee alone gives ~100% net revenue retention at best (you only ever lose, never grow, within an account). The layered models above create **negative churn** — accounts that grow without re-negotiating the base:

1. **Tier upgrades** — Starter → Team when project cap bites.
2. **Metered overages** — AI/storage creep as usage deepens.
3. **Extra workspaces** — new business unit / region.
4. **Pack→annual conversion** — the consumption on-ramp.
5. **Add-ons** — white-label, templates, premium support.

The single number to track: ARR per workspace (from `PRICING.md`), now paired with **Net Revenue Retention**. If NRR > 100%, the expansion models are working and you can grow ARR *without* adding logos. Target NRR ≥ 110% by end of year 2.

12. Implementation notes (what the code already supports)

Need	Already in schema / what to add
Plan identity	`subscriptions.plan_id` (TEXT) — use the slugs above.
Feature gating	`companies.features` JSONB + `has_feature()` / `useFeature()`.
Trial	`companies.trial_ends_at` (14-day default trigger).
Metering	`rate_limits` table + `rate_limit_check` RPC; sum overages monthly.
Billing sync	`razorpay-webhook` → `upsert_subscription()` (idempotent).
Seat count (optional)	`subscriptions.seat_count` exists but stays 0 — we do *not* price per seat.
To build	project-credit balance in `features`; overage invoicing job; Razorpay plan objects per `plan_id`; partner/parent-company billing roll-up.

13. Anti-patterns (carry-over + new)

- Never meter test records or equipment instances — corrupts field data.
- Never gate multi-tenancy, mobile, or audit *capture* — core trust.
- Don't ship a permanent free tier; the trial + free template library is the lure.
- Don't let `starter_annual` include AI/SSO — it would cannibalise Team.
- Don't price packs *below* the annual-implied rate — the upgrade must always look rational.
- Don't tag Enterprise "most popular" — Team is the workhorse.
- Don't run outcome pricing forever — migrate to flat annual by renewal 1.

Companion docs: `PRICING.md` (strategy & display), `SALES_TARGETS.md` (named accounts), financial-plan `.tex` (unit economics & 36-month ARR). Review quarterly; update slugs here when Razorpay plan objects are created.